

11	Planet X has a density of ρ and radius of R . The gravitational acceleration at its surface is a . What is the gravitational acceleration at the surface of Planet Y which has a density of 2ρ and radius of $2R$?							
	A	$0.5 a$	B	a	C	$2 a$	D	$4 a$
Answer: D $a = \frac{GM}{R^2} = \frac{G\rho V}{R^2} = \frac{G\rho(\frac{4}{3}\pi R^3)}{R^2} = \frac{4G\rho R\pi}{3}$ $a \propto \rho R$ Since ρ and R is doubled, a' is $4 a$.								

12	A particle of mass 5.0×10^{-3} kg performing simple harmonic motion of amplitude 150 mm
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